Answer key: Lesson 7, Take Away, Count Up

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Here is 68 as bars and squares. Cross out 2 bars. Then fill in the blanks.

Answer: 4 tens and 8 ones are left; 68 - 20 = 48

68 is 6 tens and 8 ones. Take away 2 tens: 4 tens and 8 ones are left. That is 48. The ones did not change.

Watch for: Crossing out 2 squares instead of 2 bars (66).; Writing 84 or 4 + 8 = 12.

S2. Subtract. Use your kit if you like.

Answer: 37, 53, 56, 81

(a) 5 tens take away 2 tens is 3 tens: 37. (b) 7 ones take away 4 ones is 3 ones: 53. (c) 8 tens take away 3 tens is 5 tens: 56. (d) 6 ones take away 5 is 1: 81.

Watch for: Changing the wrong digit (57 - 20 = 55).; Taking away from both digits.

S3. Find 68 - 25. Fill in the blanks.

Answer: 48, then 43

68 take away 20 is 48. Then 48 take away 5 is 43. So 68 - 25 = 43.

Watch for: Taking away 25 as 2 + 5 = 7 (61).; Forgetting to start part (b) from 48.

S4. Subtract. Then add to check.

Answer: 34, then 46 (the same as the start, so it checks)

46 take away 10 is 36, and 36 take away 2 is 34. Check: 34 + 12 = 46, the number we started with. So 46 - 12 = 34 is right.

Watch for: Not adding back the number taken away (adding 46 + 12).; Writing 58 for the check.

S5. Mei has 45 stickers. She gives away 13. How many stickers are left? Draw a bar for each ten and a square for each one.

Answer: 32 (4 bars and 5 squares drawn; 1 bar and 3 squares crossed out)

45 is 4 tens and 5 ones. 13 is 1 ten and 3 ones. 45 take away 10 is 35. 35 take away 3 is 32. So Mei has 32 left.

Watch for: Crossing out 13 squares.; Writing 58 (adding instead of taking away).

S6. Kumar says 59 - 20 = 57.

Answer: No. 39. He took away 2 ones instead of 2 tens (57 is 59 - 2).

59 is 5 tens and 9 ones. Take away 2 tens: 3 tens and 9 ones, which is 39. Kumar took away 2 ones, so he got 57.

Watch for: Saying he is right.; Writing 79 (adding 20).

S7. A bus has 64 seats. 21 children are sitting. How many seats are empty?

Answer: 43 seats

64 take away 20 is 44. Then 44 take away 1 is 43. So 43 seats are empty.

Watch for: Adding 64 + 21 = 85.; Taking away only 2 or only 1.

Think harder (page 3)

T1. Mei has 62 beads. Ravi has 38 beads. Find how many more beads Mei has than Ravi.

Answer: 24 more beads (38 to 40 is 2, 40 to 60 is 20, 60 to 62 is 2)

38 to 40 is 2. 40 to 60 is 20. 60 to 62 is 2. The whole gap is 2 + 20 + 2 = 24. Check: 38 + 24 = 62.

Watch for: Forgetting that two tens is 20 (writing 2).; Writing 62 - 38 as 36 or 34.; Subtracting the smaller digit from the larger in each column (62 - 38 = 36).

T2. Find 83 - 77. Show how you found it.

Answer: 6 (77 to 80 is 3, 80 to 83 is 3)

77 to 80 is 3. 80 to 83 is 3. The gap is 3 + 3 = 6. Check: 77 + 6 = 83.

Watch for: Writing only one hop (3).; Taking away 83 - 7 = 76.; Subtracting the smaller digit from the larger in each column (83 - 77 = 14).

T3. Ali has 24 stickers and Ben has 31 stickers. Then each child gets 10 more stickers. Find how many more stickers Ben has than Ali now.

Answer: 7 more (the gap stays the same: 34 and 41, and 41 - 34 = 7)

Ali now has 34 and Ben has 41. The gap from 34 to 41 is 7. It was 7 before too (24 to 31), because both children got the same 10 more.

Watch for: Writing 17 (7 + 10).; Writing 41 (Ben's number).

T4. A book has 71 pages. Ravi has read 46 pages. How many pages are left?

Answer: 25 pages (46 to 50 is 4, 50 to 70 is 20, 70 to 71 is 1)

46 to 50 is 4. 50 to 70 is 20. 70 to 71 is 1. The gap is 4 + 20 + 1 = 25. Check: 46 + 25 = 71.

Watch for: Adding 71 + 46.; Forgetting the last hop of 1 (24).; Subtracting the smaller digit from the larger in each column (71 - 46 = 35).

Go further (page 4)

G1. Fill in the blanks. Circle the quicker way for each.

Answer: 71 - 68 = 3 (count up: the numbers are close). 71 - 20 = 51 (take away: 2 tens).

71 - 68: count up. 68 to 70 is 2 and 70 to 71 is 1, so 3. The numbers are close, so the gap is short. 71 - 20: take away 2 tens to get 5 tens and 1 one, which is 51. Counting up 20 would be a long way.

Watch for: Taking away 68 from 71 one at a time.; Counting up from 20 to 71 (too long).

G2. A crate has 100 eggs. Ali sells 64 of them. How many eggs are left? Find the answer two ways.

Answer: 36 eggs. Take away: 100 - 60 = 40, then 40 - 4 = 36. Count up: 64 to 70 is 6, then 70 to 100 is 30, so 36. They match.

Take away: 100 - 60 = 40, and 40 - 4 = 36. Count up: 64 to 70 is 6, and 70 to 100 is 30, so 6 + 30 = 36. Both ways give 36. Check: 64 + 36 = 100.

Watch for: Writing 164 (adding).; Writing 40 (forgetting to take away the 4 ones).

G3. Mei says 83 - 47 and 86 - 50 have the same answer.

Answer: Yes. Both are 36. Both numbers go up by 3, so the gap does not change.

83 - 47: 47 to 50 is 3, 50 to 80 is 30, 80 to 83 is 3, so 36. 86 - 50 = 36. 83 became 86 and 47 became 50: both went up by 3, so the gap between them stayed 36.

Watch for: Finding only one answer.; Saying the answers differ because the numbers differ.

Investigation (page 5): When is counting up quicker?

Counting up is quicker when the two numbers are close together, so the gap is short (52 - 49 is 3). Taking away is quicker when the number taken away is small or a whole number of tens (52 - 10 is 42). If the numbers are far apart and the small number is awkward, both ways take longer.

Look for: A child who tries several pairs, says what they notice about how far apart the numbers are, and gives a reason. Accept any clear rule that matches examples. A child who also checks by adding is doing well.

Exit ticket (page 5)

E1. Subtract.

Answer: 34

58 take away 20 is 38, and 38 take away 4 is 34.

Watch for: Writing 82 (adding).; Writing 36 (took away 2 and 2).

E2. Fill in the blanks.

Answer: 5, then 5

57 to 60 is 3, and 60 to 62 is 2. That is 5. So 62 - 57 = 5.

Watch for: Writing only the first hop (3).; Writing 15 (62 - 47).

E3. Find 90 - 87. Circle the quicker way: take away or count up.

Answer: 3. Counting up is quicker: 87 and 90 are close together.

87 to 90 is 3, so 90 - 87 = 3. Counting up is quick because the numbers are close.

Watch for: Taking away 87 from 90 one at a time.; Writing 13.

Sneak-peek puzzles (not in the workbook)

1. 11
2. 4 more. From 58 to 60 is 2, and from 60 to 62 is 2 more. 2 and 2 make 4.